

Charles “Chaz” Merritt

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EDUCATION

University of Georgia	Athens, Georgia
<i>Master of Science (MS) in Artificial Intelligence</i>	<i>August 2025</i>
Thesis: Harnessing Reinforcement Learning for Forest Growth Modeling Under Uncertainty	GPA:3.74/4.00
University of Georgia	Athens, Georgia
<i>Bachelor of Art (BA) in Cognitive Science</i>	<i>May 2024</i>
Minor in Computer Science	GPA:3.53/4.00

RESEARCH EXPERIENCE

PERSEUS, University of Georgia	Athens, Georgia
<i>Undergraduate Researcher</i>	<i>April 2023 – May 2024</i>
Contributed to OLAF (Online Learning in Applied Forestry) by developing course materials for a module on artificial intelligence.	
PERSEUS, University of Georgia	Athens, Georgia
<i>Graduate Research Assistant</i>	<i>August 2024 – August 2025</i>
- Developed novel stand-level simulation techniques using model-free reinforcement learning and Markov decision processes. - Developed a workflow for estimating tree inventory from remotely sensed imagery using supervised fine-tuning. - Created a web interface for a Python Timber Supply optimization model using Docker and Railway.	

SELECTED PUBLICATIONS / PRESENTATIONS

- Society of American Foresters National Convention ‘25, Technical Lecture: Harnessing Reinforcement Learning for Forest Growth Modeling Under Uncertainty
- AI for Eastern US Forestry Conference ‘25: Harnessing Reinforcement Learning for Forest Growth Modeling Under Uncertainty (Pending)
- PERSEUS Annual Meeting Poster Session ‘23: Online Learning in Applied Forestry
- SOFORGIS ‘23: Deep Neural Networks enable rapid and scalable forest inventory estimates
- DOE OMNI Fire Poster Session ‘23: Exploring AI Frameworks for Next-Generation Cybersecurity with Nvidia Morpheus & SLIPS

OTHER EXPERIENCE

Northwest Management	Moscow, Idaho
<i>Geospatial Analyst</i>	<i>May – September 2024</i>
- Developed automation techniques for forest management. - Created methodology to create a quantifiable notion of ground truth for forest stand delineation.	
Oak Ridge National Laboratory	Oak Ridge, Tennessee
<i>DOE OMNI Intern</i>	<i>May 2023 – May 2024</i>
Researched network-scale intrusion prevention systems using Nvidia Morpheus and high-performance compute resources.	

HONORS AND AWARDS

PERSEUS Assistantship	<i>August 2024</i>
Promoting Economic Resilience & Sustainability in the Eastern U.S. is a large multi-institutional grant project that allows me to apply my AI & ML background to the task of solving problems in forestry.	
ISACA Foundation Digital Trust Scholarship	<i>January 2024</i>
Received a scholarship for first-generation students in cybersecurity.	

SERVICE PROJECTS

Club Archery @ UGA, President

August 2024 – May 2025

- Oversee athlete development by coaching, organizing practices, and “training the trainers.”
- Organize community service projects with Athens Area Humane Society, Golden Harvest Foodbank, and other such organizations, including acting as a liaison between Club Archery and UGA Competitive Sports.
- Register, plan for, and send club athletes to tournaments all across the country.

CERTIFICATIONS

New Media, New Media Institute – University of Georgia

May 2024

Applied Data Science, University of Georgia

May 2024